UBT309: IMMUNOTECHNOLOGY

L	T	P	Cr
3	0	0	3.0

Course Objective: To provide students with detail understanding of basic concepts of immune system, application of immunological techniques, immune dysregulation in health disorder.

Syllabus

Basic concept and cells of the Immune System: Hematopoietic stem cells, Lymphocytes, Granulocytes and Monocytes, Cell participation in innate and adaptive Immunity, MHC, Inflammatory response, Complement System

Antigens and Antibodies: Antigenicity and Immunogenicity Epitopes, Adjuvants, Superantigens, Antigen Presentation and processing, Structure and function of antibody, Antibody classes, Passive antibody therapy, Monoclonal antibody, Antibody engineering, Generation of antibody diversity

Immunological Techniques: Cross reactivity, Precipitation and Agglutination reaction, Coomb's test, RIA, ELISA, ELISPOT assay, Western blotting, Immunofluorescence and Flow cytometry

Immunological disorder: Tolerance and Autoimmunity, Types and mechanism of autoimmune diseases, Hypersensitive reactions, Different types of Hypersensitive reactions, Primary and Secondary Immunodeficiency, Tumor immunity and Tumor antigens, Transplantation types, Immunological basis of graft rejection

Vaccine and Immunotherapy: Criteria for effective vaccine, Live and Killed Vaccines, Subunit vaccines, Recombinant Vaccines, DNA vaccines, mRNA vaccines, Peptide vaccines

Course Learning Objectives (CLO)
The students will be able to:

1. Explain the role of immune cells and their mechanism in body defense mechanism.
2. Adopt immunological techniques for industrial uses.
3. Demonstrate the association of immune system with different health ailments.
4. Apply the immunological concept in developing vaccine.

Text Books

1. Punt J, Stranford S, Jones P, Owen J. Kuby- Immunology W.H. Freeman & Company (2019) VIIIth edition.

2. Murphy K., and Weaver C. Janeway Immunobiology Garland Exclusive (2016) IXth edition.

Reference Books

1. Delves P. J., Martin S. J., Burton D. R., Roitt I. M. Roitt's Essential Immunology Wiley Publisher (2017) 8th Edition.
2. Khan F.H. The Elements of Immunology, Pearson Education (2009).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	25-35
2	EST	35-45
3	Sessionals (May include Assignments/Quizzes)	30